Lab Assignment 10 — Semantic IR Using LSA and Word2Vec

IT550: Information Retrieval · Instructor: Prof. Sandip Modha

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1. Overview

This report covers a **TF-IDF** vector-space baseline and a **Latent Semantic Analysis (LSA)** retrieval system built with truncated SVD on the Cranfield collection, evaluated with P@5, MAP and nDCG@10. It also compares **Word2Vec** (CBOW and Skip-gram) word similarities with count-based (TF-IDF and PPMI) similarities. All numbers come from the notebook `202303022_lab10.ipynb`, which was run from top to bottom. Its outputs (rankings, tables, figures, `results.json`) are in `outputs/`.

Dataset and evaluation protocol

- **Cranfield**: 1,400 aerodynamics abstracts (`cran.all.1400`), 225 queries (`cran.qry`) and 1,837 judgments (`cranqrel`). The title (.T) and abstract (.W) of each document are indexed together.
- `cran.qry` uses sparse ids (001, 002, 004, ...), while `cranqrel` numbers the queries 1–225 in file order. The *i*-th query of the file is therefore query *i* in the judgments. All results use this mapping.
- **Relevant** = grade 1, 2 or 3 (for P@5 and MAP). **nDCG gain = 4 − grade** (3, 2, 1 for grades 1, 2, 3; grade 4 gives 0, and grade −1 is non-relevant).
- **222 of 225 queries** have at least one relevant document and are evaluated. Queries 31, 119 and 215 (original ids 56, 173, 339) have none and are skipped. Evaluated queries have 5.63 relevant documents on average.
- **MAP** is computed over the full ranking of all 1,400 documents. **MAP@10** (AP over the saved top-10, divided by min(|R|, 10)) is also reported. nDCG@10 uses DCG = $\Sigma\ \text{gain}_i/\log_2(i+1)$, normalised by the ideal ordering of all judged relevant documents.

2. Part 1 — Corpus preparation and TF-IDF baseline

Pipeline (identical for documents and queries): lower-casing → tokenisation with the regex `[a-z]+` → removal of scikit-learn's 318 English stop words and of 1-character tokens → Porter stemming. Example: *"what similarity laws must be obeyed when constructing aeroelastic models of heated high speed aircraft"* → `similar law obey construct aeroelast model heat high speed aircraft`.

TF-IDF: `TfidfVectorizer` with raw tf, smoothed idf = ln((1+N)/(1+df)) + 1 and L2-normalised rows, fitted on the documents only. Queries are transformed with the **same vocabulary and IDF weights**. Each query is compared with all documents by cosine similarity, and the top 10 are saved as the baseline (`outputs/rankings/tfidf_baseline_top10.csv` and `.trec`).

Statistic	Value
Documents / queries	1,400 / 225
Tokens after preprocessing	131,725 (94.1 per document on average)
Vocabulary size (stems)	4,170
Document–term matrix shape	1,400 × 4,170
Non-zero entries	77,560 (55.4 per document)
Sparsity of the document matrix	98.67 %
Query matrix shape / sparsity	225 × 4170 / 99.77 %
Query stems not in the vocabulary (ignored)	validli, trust, multhopp, stop, unnecessarili, virtu, orthodox, wildli, realiti

Table 1 — Corpus and TF-IDF matrix statistics.

3. Part 2 — Truncated SVD and the LSA representation

The TF-IDF matrix X (documents × terms) is factorised with randomized truncated SVD, $X \approx U_k \Sigma_k V_k^T$, with **k = 50** (valid, because k ≪ min(1400, 4170)). Documents are represented by $XV_k = U_k\Sigma_k$. A query vector q is folded into the **same space** as qV_k. Documents are ranked by cosine similarity in the latent space, and the top 10 are saved (`lsa_k50_top10.csv`).

Quantity (k = 50)	Value
Explained variance ratio of the 50 retained components	0.3115 (31.15 %)
Share of $\|X\|^2_F$ kept ($\Sigma\sigma_i^2$ / $\|X\|^2_F$)	0.3488
Document matrix in latent space	1,400 × 50
Query matrix in latent space	225 × 50
Term loadings V_k	4,170 × 50

Table 2 — LSA representation with k = 50.

The variance is spread over many components. The first 50 components explain only 31 % of the variance, and even 300 explain 70 % (Figure 1), which is typical for a sparse, high-rank TF-IDF matrix of short abstracts. The first component has a lower ratio than the second because TruncatedSVD does not centre the data, so component 1 mostly captures the mean direction. The leading components are interpretable topics. Component 2 contrasts *wing/buckling/flutter* with *boundary layer/heat transfer*, and component 3 groups *buckling, shell, stress, cylinder, load*.

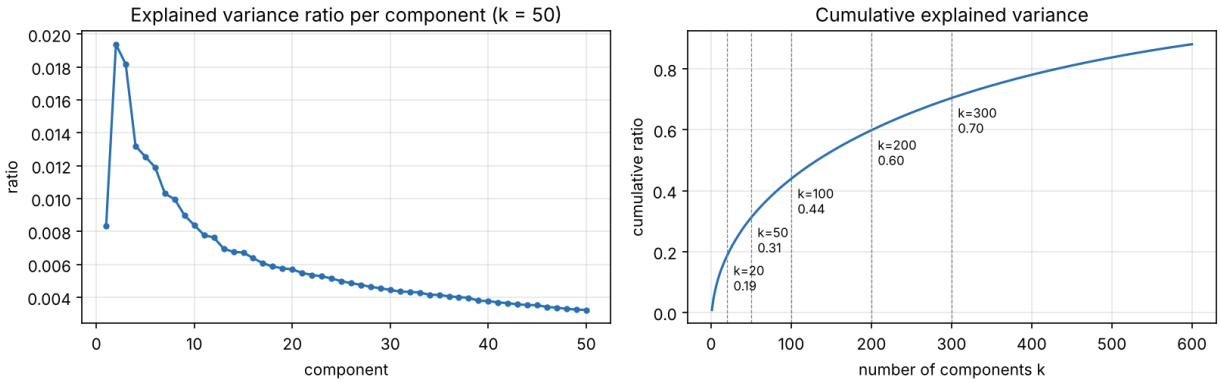


Figure 1 — Explained variance per component (left) and cumulative explained variance (right).

4. Part 3 — Evaluation and dimensionality analysis

4.1 TF-IDF vs LSA (k = 50)

System (222 queries)	P@5	MAP	MAP@10	nDCG@10
TF-IDF baseline	0.2739	0.2912	0.2441	0.3495
LSA, k = 50	0.2279	0.2685	0.2186	0.3067
Relative change	-16.8 %	-7.8 %	-10.4 %	-12.3 %

Table 3 — Retrieval quality of the baseline and of LSA with k = 50.

Top-10 comparison. The two systems share on average only **5.23** of their 10 documents (Jaccard 0.377), so LSA does more than re-order the TF-IDF ranking. LSA's top-10 lists contain **101** relevant documents that TF-IDF misses, but they lose **133** that TF-IDF finds. In total LSA has 407 relevant documents in its top-10 lists against 439 for TF-IDF, and 52 queries without any relevant document against 39.

Per query, LSA (k = 50) has a higher AP on 92 queries and a lower AP on 124 (6 ties). nDCG@10 is higher on 78 queries, lower on 108, with 36 ties (Figure 2).

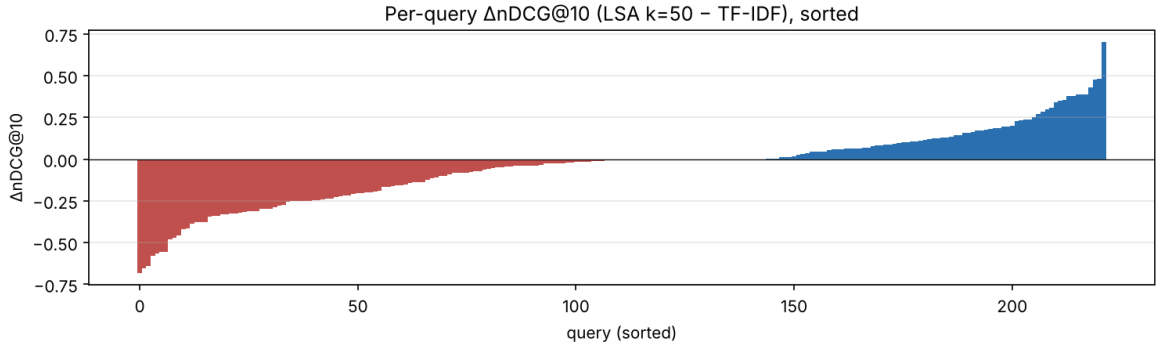


Figure 2 — Per-query difference in nDCG@10 (LSA k = 50 minus TF-IDF), sorted.

4.2 Queries where LSA improves or worsens the ranking

qid	Query	AP TF-IDF	AP LSA	nDCG TF-IDF	nDCG LSA	ΔnDCG@10
184	work on small-oscillation re-entry motions .	0.095	0.601	0.000	0.702	+0.702
167	exact solution methods for calculating the ablative mass loss of a material ablating	0.086	0.375	0.000	0.480	+0.480
180	how does scale height vary with altitude in an atmosphere .	0.204	0.527	0.354	0.831	+0.477
165	are the stable profiles of a compressible boundary layer induced by a moving wave kno	0.361	1.000	0.571	1.000	+0.429
53	what investigations have been made of the flow field about a body moving through a ra	0.049	0.274	0.000	0.387	+0.387
95	what is the theoretical heat transfer distribution around a hemisphere .	0.077	0.200	0.000	0.387	+0.387

Table 4 — Largest improvements by LSA (k = 50).

qid	Query	AP TF-IDF	AP LSA	nDCG TF-IDF	nDCG LSA	ΔnDCG@10
136	what are the results for the creep buckling of round tubes under external pressure .	1.000	0.125	1.000	0.316	-0.684
148	papers on small deflection theory for buckling of sandwich cylinders .	0.439	0.070	0.655	0.000	-0.655
198	is there any information available on the difference in the effects of various edge c	0.483	0.090	0.639	0.000	-0.639
5	what chemical kinetic system is applicable to hypersonic aerodynamic problems .	0.420	0.080	0.582	0.000	-0.582
24	what are the factors which influence the time required to invert large structural mat	0.479	0.122	0.721	0.154	-0.567
64	can't the static deflection shapes be used in predicting flutter in place of vibratio	0.643	0.216	0.860	0.303	-0.557

Table 5 — Largest degradations by LSA (k = 50).

Improvement — qid 184 “work on small-oscillation re-entry motions” (nDCG@10 0.00 → 0.70, AP 0.10 → 0.60). The relevant documents use different wording for the same topic: *“dynamic stability of vehicles traversing ascending or descending paths”* (doc 67), *“oscillatory motion of manned vehicles **entering** the earth's atmosphere”* (doc 716), *“oscillations of a vehicle **entering** a planetary atmosphere”* (doc 499). The query stem `entri` (from *re-entry*) never matches `enter`, and the generic words *work* and *small* add noise, so TF-IDF ranks these documents 13th–94th. In the latent space *missile, vehicle, descending, atmosphere, oscillation, motion* load on the same dimensions. LSA moves doc 67 from rank 57 to 1 and doc 716 from 20 to 2. This is the **synonymy / vocabulary-mismatch** problem that LSA is designed to solve.

Degradation — qid 136 “creep buckling of round tubes under external pressure” (AP 1.00 → 0.13). The single relevant document (1021) shares five stems with the query, and TF-IDF ranks it first. The collection holds a dense cluster of *creep buckling* papers (950, 951, 1013, 1017, 1031, ...). With 50 dimensions they collapse onto nearly the same latent vector, and the words that single out the relevant paper (*tubes, round, external pressure*) are averaged away, so doc 1021 drops to rank 8. Queries 148 (*sandwich* cylinders), 5 (*chemical kinetic* systems) and 9 (*slip flow* heat transfer) fail the same way: a rare, discriminative term is what matters, and the truncated SVD discards it. This is the **precision cost** of merging related topics.

Share of query stems in the relevant document	Pairs	Median rank TF-IDF	Median rank LSA	LSA ranks it higher
0-20 %	237	397	188	72.6 %
20-40 %	413	30	34	49.9 %
40-60 %	387	13	16	39.8 %
60-100 %	212	4	6.5	28.8 %

Table 6 — All (query, relevant document) pairs grouped by literal word overlap with the query.

Table 6 supports this explanation across all queries. When a relevant document shares at most 20 % of the query's stems, LSA ranks it higher than TF-IDF in 72.6 % of cases (median rank 397 → 188).

When it shares more than 60 %, TF-IDF wins and LSA ranks it higher only 28.8 % of the time. **LSA helps on vocabulary mismatch and hurts where exact matching already works.**

4.3 Effect of the number of latent dimensions and runtime

System	k	Explained var.	P@5	MAP	MAP@10	nDCG@10	Index / fit time (s)	Query time (s, 225 q.)
TF-IDF	4,170 (vocab.)	—	0.2739	0.2912	0.2441	0.3495	0.163	0.026
LSA	20	18.7 %	0.1703	0.2071	0.1596	0.2340	0.122	0.058
LSA	50	31.2 %	0.2279	0.2685	0.2186	0.3067	0.205	0.037
LSA	100	43.9 %	0.2658	0.3016	0.2509	0.3531	0.447	0.036
LSA	200	59.8 %	0.2856	0.3187	0.2738	0.3817	0.785	0.036
LSA	300	70.4 %	0.2811	0.3127	0.2634	0.3706	1.245	0.032

Table 7 — Retained dimensions, retrieval metrics and runtime (median of 3 runs). TF-IDF fit time includes preprocessing and building the matrix, which LSA also needs before its SVD.

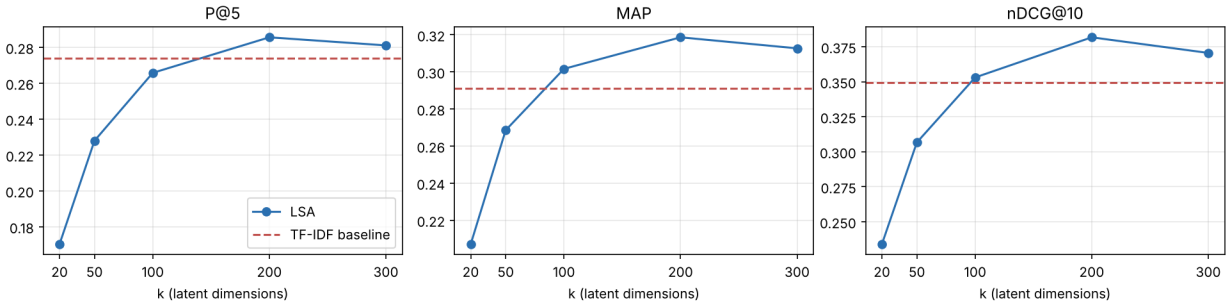


Figure 3 — Retrieval quality of LSA as a function of k (dashed line: TF-IDF baseline).

- Quality **rises with k and then falls slightly**. With k = 20 the space is too coarse (P@5 0.170). k = 50 is below the baseline on every metric, and k = 100 roughly ties it (MAP +3.6 %, nDCG@10 +1.0 %, P@5 -3.0 %).
- **k = 200 is best**: P@5 0.2856 (+4.3 %), MAP 0.3187 (+9.4 %), nDCG@10 0.3817 (+9.2 %). At k = 300 the scores drop again (MAP 0.3127) because LSA moves back toward TF-IDF as k approaches the rank of the matrix. For this small, technical collection the k = 50 suggested in the handout is too aggressive.
- Explained variance (19 % → 70 %) is a poor guide for choosing k. The best k keeps only 60 % of the variance, so k should be tuned on retrieval metrics.
- The randomized SVD is stable: over 5 seeds the k = 50 metrics vary by at most ±0.003.
- **Runtime:** the SVD (indexing) cost grows roughly linearly with k (0.12 s at k = 20, 1.25 s at k = 300, on top of building the TF-IDF matrix). Query processing (folding in 225 queries, scoring and sorting 1,400 documents) takes 26–58 ms for every system, well under 1 ms per query. At this scale it is dominated by sorting and hardly depends on k.

5. Part 4 — Word embeddings with Word2Vec

Training data: the same 1,400 documents (title + abstract), split into 11,413 sentences (131,618 tokens). The tokenizer and stop-word list are the same as in Part 1. Stemming is not applied here, so neighbours stay readable. **Settings** (identical except the architecture): vector size 100, window 5, min_count 5 (vocabulary of 2,697 words), negative sampling with 10 noise words, sub-sampling 1e-3, 30 epochs, one worker thread with a fixed seed and hash function for reproducibility. **CBOW** (sg = 0) predicts a word from the average of its context. **Skip-gram** (sg = 1) predicts each context word from the word.

5.1 Five nearest words — CBOW vs Skip-gram

Term (freq.)	CBOW top-5 (cosine)	Skip-gram top-5 (cosine)	Shared
supersonic (629)	transonic (0.57), subsonic (0.54), traveling (0.53), uncambered (0.51), wakes (0.49)	subsonic (0.56), overexpanded (0.54), intakes (0.53), spiked (0.52), uncambered (0.51)	2
turbulent (353)	separated (0.52), turbulence (0.50), laminar (0.48), zone (0.46), boundary (0.40)	laminar (0.57), boundary (0.54), layer (0.51), turbulence (0.51), stabilization (0.51)	3

Term (freq.)	CBOW top-5 (cosine)	Skip-gram top-5 (cosine)	Shared
wing (641)	wings (0.68), cropped (0.61), tail (0.60), arrangements (0.59), flaps (0.56)	wings (0.62), cropped (0.58), tail (0.58), tilt (0.58), tilting (0.57)	3
heat (749)	mass (0.47), isothermal (0.46), heating (0.46), shield (0.44), diffusion (0.43)	transfer (0.79), shield (0.62), refractory (0.58), transpiration (0.54), protection (0.52)	1
shock (794)	bow (0.60), originating (0.55), detached (0.55), weak (0.52), blast (0.52)	wave (0.67), attenuation (0.57), incident (0.57), driver (0.56), striking (0.56)	0
buckling (371)	deformations (0.64), wrinkling (0.60), buckle (0.53), columns (0.53), vessels (0.52)	rings (0.61), unstiffened (0.60), postbuckling (0.58), unpressurized (0.58), stiffened (0.58)	0
aerofoil (35)	airfoil (0.49), wings (0.48), discs (0.48), delta (0.47), tapered (0.47)	inverted (0.52), discs (0.52), split (0.52), tapered (0.50), mapping (0.50)	2
nozzle (200)	exit (0.64), nozzles (0.60), convergent (0.59), exhaust (0.56), converging (0.52)	exit (0.65), laval (0.58), nozzles (0.57), throat (0.55), choked (0.54)	2

Table 8 — Five nearest neighbours by cosine similarity.

Model	Mean cos, 12 related pairs	Mean cos, 500 random pairs	Gap	Training time (s)
CBOW	0.465	0.059	0.405	11.4
Skip-gram	0.491	0.198	0.294	36.4

Table 9 — Comparison of the two configurations.

- **Kind of neighbours.** CBOW returns words that can **replace** the target in a sentence: *supersonic → transonic, subsonic*; *buckling → deformations, wrinkling, buckle*; *aerofoil → airfoil, wings, delta*. Skip-gram returns more **topical associates**, including rare, specific words: *heat → transfer, shield, refractory, transpiration* (the heat-shield topic); *buckling → rings, unstiffened, stiffened, postbuckling*.
- **Agreement** is low. On average only 21 % of the top-10 neighbours are shared (Jaccard 0.207, 1,000 most frequent words).
- Both models separate related from random pairs. CBOW has the larger gap. Skip-gram gives even random pairs a cosine of about 0.2, so its scores must be read relative to that baseline. Skip-gram trains 3.2× slower, because it makes one prediction per (word, context word) pair.

5.2 Comparison with count-based similarity

Two count-based word representations are built on the same unstemmed vocabulary. **TF-IDF (document level)**: each word is a column of the document–term TF-IDF matrix, so words are similar when they occur in the *same documents*. **PPMI (window level)**: word × context counts within ±5 words, weighted by positive pointwise mutual information. All similarities are cosines.

Term	TF-IDF (same documents)	PPMI (±5-word window)	Skip-gram
supersonic	flow (0.34), speeds (0.31), mach (0.30), theory (0.28), pressure (0.25)	subsonic (0.33), speeds (0.31), wings (0.28), flow (0.28), bodies (0.28)	subsonic (0.56), overexpanded (0.54), intakes (0.53), spiked (0.52), uncambered (0.51)
turbulent	boundary (0.35), layer (0.34), layers (0.28), friction (0.26), laminar (0.25)	laminar (0.37), boundary (0.35), layers (0.32), layer (0.31), transfer (0.24)	laminar (0.57), boundary (0.54), layer (0.51), turbulence (0.51), stabilization (0.51)
wing	wings (0.30), aspect (0.30), lift (0.30), combinations (0.30), body (0.29)	wings (0.43), tail (0.35), chord (0.33), trailing (0.31), flaps (0.31)	wings (0.62), cropped (0.58), tail (0.58), tilt (0.58), tilting (0.57)
heat	transfer (0.83), temperature (0.40), conduction (0.39), laminar (0.35), stagnation (0.30)	transfer (0.69), temperature (0.34), conduction (0.32), laminar (0.30), surface (0.28)	transfer (0.79), shield (0.62), refractory (0.58), transpiration (0.54), protection (0.52)
shock	wave (0.54), waves (0.47), interaction (0.29), detached (0.28), driver (0.27)	wave (0.55), waves (0.43), tube (0.29), reflected (0.28), flow (0.27)	wave (0.67), attenuation (0.57), incident (0.57), driver (0.56), striking (0.56)
buckling	shells (0.45), columns (0.41), axial (0.38), compression (0.36), creep (0.35)	shells (0.50), cylindrical (0.39), cylinders (0.37), plates (0.35), plastic (0.34)	rings (0.61), unstiffened (0.60), postbuckling (0.58), unpressurized (0.58), stiffened (0.58)
aerofoil	split (0.30), calculating (0.26), ground (0.24), formulae (0.24), aerofoils (0.21)	split (0.19), calculate (0.18), rae (0.18), circle (0.18), deg (0.18)	inverted (0.52), discs (0.52), split (0.52), tapered (0.50), mapping (0.50)
nozzle	nozzles (0.52), exit (0.32), expanded (0.32), convergent (0.30), throat (0.30)	exit (0.37), jet (0.31), nozzles (0.28), throat (0.28), diameters (0.26)	exit (0.65), laval (0.58), nozzles (0.57), throat (0.55), choked (0.54)

Table 10 — Five nearest neighbours under count-based and embedding similarity.

Word pair	df(a)	df(b)	Docs with both	TF-IDF	PPMI	LSA k=100	CBOW	Skip-gram
supersonic – hypersonic	267	170	28	0.086	0.182	0.088	0.467	0.409
laminar – turbulent	233	127	70	0.247	0.369	0.258	0.485	0.566
aerofoil – airfoil	18	53	1	0.013	0.147	-0.032	0.488	0.286
wing – aerofoil	181	18	4	0.035	0.133	0.080	0.456	0.380
temperature – heat	250	254	138	0.399	0.341	0.511	0.382	0.473
cylinder – cone	131	92	22	0.142	0.349	0.088	0.384	0.424
buckling – shells	120	85	51	0.449	0.505	0.427	0.444	0.573
shock – wave	237	168	117	0.538	0.547	0.719	0.345	0.674
nozzle – jet	71	92	26	0.267	0.308	0.246	0.475	0.479
viscous – inviscid	122	79	28	0.274	0.210	0.558	0.546	0.471
lift – drag	134	136	58	0.415	0.478	0.530	0.449	0.611
plate – plates	184	72	29	0.153	0.295	1.000	0.653	0.552

Table 11 — Cosine similarity of selected word pairs (LSA uses stems, so plate/plates = 1.0).

Spearman ρ	TF-IDF	PPMI	CBOW	Skip-gram
TF-IDF	1.000	0.664	0.368	0.403
PPMI	0.664	1.000	0.542	0.515
CBOW	0.368	0.542	1.000	0.662
Skip-gram	0.403	0.515	0.662	1.000

Table 12 — Rank correlation of pairwise similarities over all 44,850 pairs of the 300 most frequent words.

- Document-level **TF-IDF** neighbours are topical associates (*supersonic* → *flow, speeds, mach, theory, pressure*). A pair scores high only when the words occur in the same abstracts (*heat–transfer* 0.83, *shock–wave* 0.54).
- **PPMI** uses local context and is already closer to Word2Vec (*supersonic* → *subsonic*; *wing* → *wings, tail, chord, flaps*). Word2Vec correlates more with PPMI (ρ ≈ 0.52–0.54) than with TF-IDF (ρ ≈ 0.37–0.40). This is consistent with skip-gram with negative sampling implicitly factorising a shifted PMI matrix.
- Words that **substitute** for each other but rarely appear in the same abstract get almost zero TF-IDF similarity (*aerofoil–airfoil* 0.013, *wing–aerofoil* 0.035, *supersonic–hypersonic* 0.086). CBOW gives them 0.46–0.49. LSA term vectors behave like smoothed TF-IDF: they help when the documents themselves overlap (*viscous–inviscid* 0.27 → 0.56) but not for disjoint spelling variants.

5.3 A relationship that embeddings capture and word overlap misses: aerofoil / airfoil

The abstracts use both the British spelling **aerofoil** (18 documents) and the American **airfoil** (53 documents) for the same object, and only **one** document contains both. Exact matching treats them as unrelated: the stems differ, so a query that says *airfoil* gets no credit from a document that says *aerofoil*. Their document-level TF-IDF cosine is **0.013**. LSA (k = 100) gives **−0.03**, because the two spellings occur in disjoint sets of documents.

CBOW ranks airfoil as the nearest neighbour of aerofoil (rank 1 of 2,696 words, cosine 0.49). Word2Vec does not need the two words to co-occur. It needs them to occur in **similar contexts**, and both appear next to *theory, pressure, lift, distribution, section, mach, velocity, transonic, flow*. Because CBOW learns to predict a word from its context, two words that fill the same slot get similar vectors. This second-order (distributional) relation is invisible to first-order word overlap. A systematic search found more such pairs that never share a document but have Skip-gram cosine ≈ 0.55, for example *diameter–width* and *integration–solving*. In retrieval this affects queries 42 and 43 (“transonic flow around an *airfoil*”): relevant doc 468 says only *aerofoil* and is found (ranks 6 and 12) only through other shared words. Expanding queries with embedding neighbours would close this gap.

Skip-gram ranks *airfoil* only 450th for *aerofoil*. With 35 occurrences, Skip-gram fits the word's specific co-occurring terms (*split, discs, inverted*) rather than its general slot, so CBOW's context averaging is more robust on this small corpus. Note that Word2Vec vectors are *static*, one per word type learnt from its contexts. Fully contextual models such as BERT would also distinguish senses of a word from

sentence to sentence.

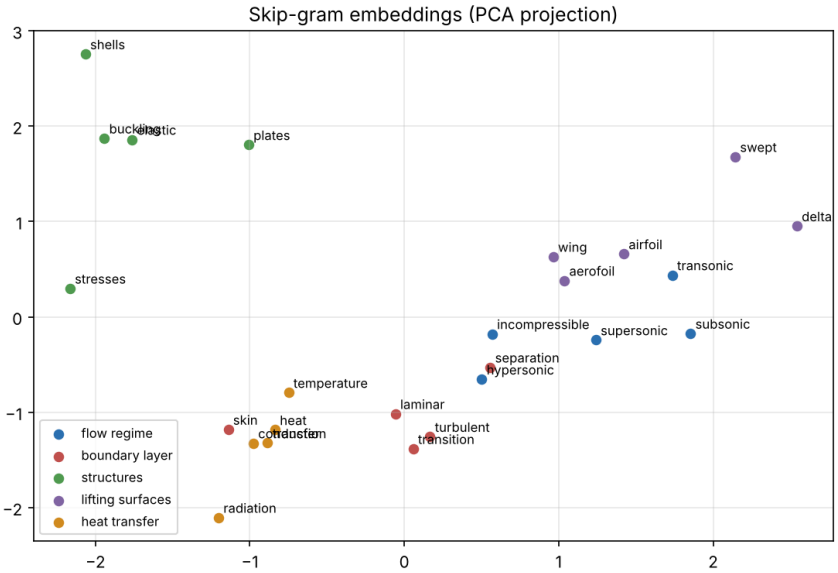


Figure 4 — 2-D PCA projection of Skip-gram vectors for five semantic groups. The groups form separate regions, and aerofoil/airfoil/wing lie together.

6. Conclusions

- **TF-IDF baseline** (4,170 stems, 98.7 % sparse): P@5 0.2739, MAP 0.2912, nDCG@10 0.3495 on 222 queries.
- **LSA with k = 50** keeps 31 % of the variance and is **worse** than TF-IDF on Cranfield (MAP 0.2685, nDCG@10 0.3067). It recovers relevant documents written with different vocabulary (qid 184) but blurs precise queries whose key term is rare (qid 136).
- **Dimensionality matters most**: k ≈ 200 gives the best system (MAP 0.3187, nDCG@10 0.3817, about +9 % over TF-IDF). Small k under-fits, and very large k turns LSA back into TF-IDF. Indexing cost grows with k, while query cost stays well under 1 ms per query.
- **Word2Vec**: CBOW gives substitutable neighbours and Skip-gram gives topical associates. Both agree more with window-based PPMI than with document-level TF-IDF, and they capture distributional relations such as *aerofoil* ≈ *airfoil* that word-overlap similarity cannot see.

7. Submission contents and reproduction

File / folder	Content
202303022_lab10.ipynb	Notebook with all code and visible outputs (Parts 1–4)
202303022_lab10.html	The executed notebook exported as HTML (easy to view)
202303022_lab10_report.pdf	This report
outputs/rankings/	Top-10 rankings for every query: TF-IDF baseline and LSA k = 20, 50, 100, 200, 300 (CSV and TREC format)
outputs/tables/, outputs/figures/	Per-query metrics, metrics by k, neighbour tables, word-pair similarities; all figures
outputs/results.json	All reported numbers in machine-readable form
data/	Cranfield files (cran.all.1400, cran.qry, cranqrel)
make_report.py, requirements.txt, README.md	Report generator, dependencies, instructions

To reproduce: `pip install -r requirements.txt`, then `jupyter nbconvert --to notebook --execute --inplace 202303022_lab10.ipynb` and `python make_report.py`. All random components (randomized SVD, Word2Vec) use fixed seeds. Runtimes depend on the machine.